SARVESH BASKAR

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SUMMARY

Dual-degree CS and Physics graduate from BITS Pilani with extensive research experience in Large Language Models (LLMs), Vision-Language Models (VLMs), and agentic reasoning at UMD and UMBC. Co-author of 6 peer-reviewed publications (EMNLP Findings, ACL Findings, NAACL, COLM Workshop). My research interests span Multimodal AI, Vision-Language Models, LLM Agents, Reasoning & Planning, Visual Grounding, Embodied AI, Robotics, Computer Vision, and Vision-Language-Action Models.

EDUCATION

Birla Institute of Technology and Science (BITS) Pilani Sep 2020 to July 2025
Dual Degree:
Bachelor of Engineering in Computer Science Cumulative GPA: 8.46/10.00
Master of Science in Physics
Thesis Title: An Approach to Resolving the Persona Knowledge Gap in LLMs during Multi-Turn Conversations

SKILLS

Programming: Python, C/C++, Bash, SQL, LaTeX
ML/AI: PyTorch, TensorFlow, OpenCV, Transformers, vLLM, LangChain, LlamaIndex, DSPy, RAG, VLMs
Systems: Git, Linux, Docker, Slurm, Redis, ChromaDB, FAISS, Ollama, MCP, HuggingFace

PUBLICATIONS

Peer-Reviewed Publications

- Suri, Manan, **Sarvesh Baskar**, and Dinesh Manocha. "Frames2LoRA: Parametric Video Internalization for Vision-Language Models."
  EMNLP Findings, 2026.
- **Baskar, Sarvesh**, Raviteja Bommireddy, Saksham Kumar Sharma, Harsha Kokel, Manas Gaur, Michael Katz, Shirin Sohrabi, and Kavitha Srinivas. "PlanForge: Synthesizing Planners from Natural Language with an LLM-Driven Architect–Builder–Runner Pipeline."
  COLM Workshop on Efficient Reasoning, 2026.
- Agrawal, Aakriti, Gouthaman KV, Rohith Aralikatti, Gauri Jagatap, Jiaxin Yuan, **Sarvesh Baskar**, Vijay Kamarshi, Andrea Fanelli, and Furong Huang. "Towards Mitigating Hallucinations in Large Vision-Language Models by Refining Textual Embeddings."
  ACL Findings, 2026.
- **Baskar, Sarvesh**, Manas Gaur, Srinivasan Parthasarathy, and Tanmay Tulsidas Verlekar. "(CPER) From Guessing to Asking: An Approach to Resolving Persona Knowledge Gap in LLMs during Multi-Turn Conversations."
  NAACL SRW, 2025.
- **Baskar, Sarvesh**, Muhammad R. Islam, Zikui Cai, Ankit Nakhawa, Anirudh Satheesh, Tom Goldstein, and Furong Huang. "The Low-Frequency Trap: Why Scaling Doesn't Solve Simple Temporal Counting."
  ICLR 2026 Workshop on I Can't Believe It's Not Better, 2026.
- Mohseni, Seyedreza, **Sarvesh Baskar**, Arman Hossain, Aritran Piplai, Edward Raff, and Manas Gaur. "Analyzing Chain of Thought (CoT) Approaches in Control Flow Code Deobfuscation Tasks."
  WoRMA, 2026.

Preprints

- **Baskar, Sarvesh***, Zikui Cai*, Shayan Shabihi*, Anirudh Satheesh, Muhammad R. Islam, Udari Madhushani Sehwag, Tom Goldstein, and Furong Huang. "The Low-Frequency Trap: Video–Language Models Fail at Simple Event Bookkeeping."
  arXiv preprint, 2026.

ACADEMIC EXPERIENCE

University of Maryland, College Park — Remote
Research Assistant (Furong Huang's Lab) — Jul 2025 - Present

- Developed trace-grounded parametric profiling to evaluate temporal event bookkeeping in frontier VLMs, uncovering the Low-Frequency Trap (*ICLR 2026 Workshop*).
- Implemented VisAlign's visual-context refinement of text embeddings in OpenQwen2VL and ran pretraining and evaluation pipelines to mitigate hallucinations (*ACL Findings 2026*).
- Collaborated with PhD students and postdoctoral researchers to design grounding interventions, execute controlled Slurm-based ablations, and track metrics for reproducibility.
- Studied relevant literature to inform methodology and prepared publication-ready figures, tables, and manuscript sections.

University of Maryland, Baltimore County (UMBC) — MD, USA
Research Assistant (KAI2 Lab) — Advisor: Dr. Manas Gaur — Aug 2024 - Jul 2025

- Led experiments, analysis, and writing for a first-author NAACL 2025 SRW paper on conversation preference elicitation for LLMs.
- Collaborated with PhD students to scope tasks and plan experiments on persona knowledge gap resolution and code reasoning.
- Lead developer for PlanForge, which extends AutoTOS to harder ACPBench classical planning domains in collaboration with IBM where I defined experiments, added benchmarks, and standardized analysis.
- Developed LLM-driven planning systems, code deobfuscation modules, and evaluation pipelines using Python.
- Co-wrote sections of the Samsung START 2025 proposal, including scope, methodology, milestones, and expected outcomes.

TCS Research - Datalab, APPCAIR, BITS Pilani — Goa, India
Student Researcher (APPCAIR) — Jan 2024 - May 2024

- Conducted comprehensive literature reviews to scope tasks, identify baselines, datasets, and evaluation protocols for automated planning and code synthesis.
- Translated high level research goals into concrete experimental plans and weekly milestones with professors and industry collaborators.
- Implemented discussed methodology pipelines in Python with version control, unit tests, and clear documentation, achieving +30% code yield through iterative prompt repair.
- Prepared weekly written reports and presentations that summarized results, insights, and next steps for faculty and industry stakeholders.
- Coordinated code handoff by organizing repositories, packaging notebooks for maintainability, and contributed to manuscript sections on related work and methods.

Birla Institute of Technology and Science (BITS) Pilani — Goa, India
Graduate Teaching Assistant — Aug 2023 - May 2024
BITS Pilani, Goa Campus (Department of Physics) — *Goa, India*

- Assisted instruction for the graduate-level course *Mathematical Methods of Physics* (Fall 2023-24).
- Graded assignments and problem sets
- Conducted demonstrations for mathematical methods, including integration and differential equations.
- Supported exam evaluation; provided feedback to students and instructors to improve learning outcomes.

SELECTED RESEARCH PROJECTS

The Low-Frequency Trap: Event Bookkeeping in VLMs (UMD) *Sep 2025 - Jul 2026*

- **Problem:** Standard video VLM benchmarks entangle event count, rate, and visual clutter, scoring only final answers without auditing whether intermediate temporal reasoning supports the final count.
- **What I did:** Developed a trace-grounded parametric profiling framework generating 2,190 controlled videos with executable ground-truth traces across event count N and frequency F grid sweeps. Audited model-reported timestamps (trace F_1, precision, recall, ACR, RFR) under FPS sampling, oracle keyframe, and prompting interventions.
- **Impact and Results:** Exposed a staged temporal failure where Gemini 3.6 Flash recovers only 18.1% of true events under high temporal load, demonstrating that denser frame sampling can improve final-count accuracy without improving event-trace fidelity.

PlanForge: Executable Planner Generation from Natural Language (UMBC) *May 2025 - Present*

- **Problem:** Traditional direct LLM plan generation from natural language is highly brittle, unverified, and generalizes poorly to multi-constraint tasks with multiple valid solutions.
- **What I built:** An Architect–Builder–Runner framework that converts natural-language planning problems into executable solver programs rather than direct textual plans. Designed an Architect stage for state/logic inference, a Builder pipeline for solver code generation/repair, and a deterministic Runner to verify outputs.
- **Impact and Results:** Evaluated PlanForge on ACPBench, Tower of Hanoi, Frogs Jumping, and NaturalPlan, achieving a 100% success rate on NaturalPlan trip planning, meeting planning, and calendar scheduling; accepted at COLM 2026 Workshop on Efficient Reasoning.

Frames2LoRA: Zero-Token Video Question Answering (UMD) *Mar 2026 - May 2026*

- **Problem:** Video reasoning in VLMs is extremely token-heavy, causing high latency, large context window usage, and context degeneration during long-video question answering.
- **What I built:** A perceiver hypernetwork running alongside a VLM to predict adapter weights in a single forward pass, eliminating visual tokens from query context and enabling composition of independent video chunk adapters in rank space.
- **Impact and Results:** Reduces query-time visual tokens by 1500x and TTFT by up to 80x while maintaining accuracy; accepted at EMNLP Findings 2026.

VisAlign: Mitigating Hallucinations via Refining Textual Embeddings (UMD) *Oct 2025 - Jan 2026*

- **Problem:** Large vision language models tend to over rely on language tokens because visual embeddings are simply appended to the text stream, which leads to weak visual grounding and hallucination.
- **What I did:** Implemented VisAlign's visual-context refinement of text embeddings in OpenQwen2VL and ran the pretraining and evaluation pipeline.
- **Impact and Results:** Demonstrated lower hallucination rates with improved visual grounding across multi-modal benchmarks; published at ACL Findings 2026.

Multi-Step Reasoning using Process Reward Models (UMD) *Jul 2025 - Aug 2025*

- **Problem:** Outcome-only supervision in math reasoning rewards the final answer but ignores the quality of intermediate steps, leading to brittle reasoning chains.
- **What I built:** An end-to-end PRM pipeline that generates multi-step trajectories, assigns weak step-level labels using verifier/self-consistency signals, and uses that score at inference for PRM-guided answers.
- **Impact and Results:** Majority voting was consistently strongest with a 92% accuracy compared to PRM with a 68% accuracy on GSM8K.

INDUSTRY EXPERIENCES

Avyott Goa, India / Remote
AI Consultant Aug 2025 - Present

- Designed and shipped a multimodal RAG chatbot embedded inside client products, able to process text, images, and tables for grounded and cited answers.
- Wrote and deployed Model Context Protocol (MCP) servers to connect with multiple ticketing systems, enabling chat driven ticket creation, updates, and status sync.
- Implemented a clarification step that asks targeted questions before answering, based on my thesis research, which improved personalization and intent capture.
- Deployed the product for enterprise use at IPCA and Taj, integrating with internal knowledge bases and existing workflows.

Techisy Goa, India / Remote
Summer Intern May 2024 - July 2024

- Built an AI-powered duplicate bill identification system (95% accuracy) combining OCR, OpenAI embeddings, and CLIP, reducing manual review time.
- Created a RAG-based resume matching solution using LangChain, Ollama (Llama 3), and FAISS to align resumes with job descriptions.
- Deployed prototypes and iterated with stakeholders to improve screening efficiency.

ENGLISH TESTS

IELTS (Academic): Overall: 7.5 | Listening: 7.5 | Reading: 7.5 | Speaking: 8.0 | Writing: 7.0